YEARS 3 TO 13

MATHS FROM AGE 7 TO A-LEVEL

quarter™

# The *curriculum*

Everything your child can learn on quarter, year by year, with questions to ask at home.

STEM LAB

CODE, DATA, AI AND QUANTUM IDEAS

A reference for parents and carers

quarter™ is a trading name of Wristed Ltd

WELCOME

# Welcome

This booklet is a map of everything quarter covers, so you can see where your child is, what comes next and how to help at home.

## How to use it

- Find your child's school year in the chapters that follow.
- Read what each topic covers, in plain words.
- Use the “Ask” question beside a topic to start a conversation about it.

## What you'll find

- Maths from Year 3 to A-level, topic by topic.
- The 11+ strand and practice papers.
- The STEM lab: computing, data, AI and quantum ideas.
- The study library, tools and the parents' area.

SHORT, REGULAR PRACTICE BEATS LONG SESSIONS. TEN MINUTES MOST DAYS IS PLENTY.

## The idea behind quarter

Every question works out its own answer, so hints, worked steps and feedback always match the question. Topics come back later, a little at a time, so what's learned stays learned. Children are never rushed on normal practice, and the only timed activity is the clearly labelled 11+ practice paper.

USING QUARTER

# How quarter works

1. **A short check-up finds the starting point**
   About 24 questions that adapt to your child, so the first plan fits them, not just their school year.
2. **A weekly plan keeps things steady**
   Each week focuses on a few topics, with review of earlier ones mixed in.
3. **Practice is short and kind**
   A hint is always there, “Show the steps” explains a worked answer, and an optional AI tutor can talk a problem through without giving the answer away.
4. **Working-out tools help on paper and on screen**
   Counters, number lines, grids and method sheets are a tap away, and the keyboard works everywhere.
5. **Topics return at the right moment**
   Anything that was tricky comes back sooner. Anything secure comes back later.
6. **You see the progress**
   The parents' area shows what's secure, what needs attention, and a progress report you can print or save for any dates you choose.

## What the status words mean

- **New**: not tried yet.
- **Learning**: being worked on.
- **Getting there**: mostly right, still wobbly.
- **Got it**: secure, and brought back later to stay that way.

## Where it works

In any modern browser on phones, tablets and computers. Progress is saved to your account, so a child can carry on from any device.

INSIDE

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THE WHOLE PICTURE

# At a glance

**199**
maths topics

**21**
11+ reasoning topics

**29**
STEM lab topics

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illustrated study guides

## Key Stage 2

Years 3 to 6, ages 7 to 11

**87** topics

Number · Ratio and proportion · Algebra · Measures · Geometry · Statistics

## Key Stage 3

Years 7 to 9, ages 11 to 14

**40** topics

Number · Ratio and proportion · Algebra · Measures · Geometry · Statistics

## GCSE

Years 10 to 11, ages 14 to 16

**37** topics

Number · Ratio and proportion · Algebra · Measures · Geometry · Statistics

## A-level

Years 12 to 13, ages 16 to 18

**35** topics

Algebra · Measures · Geometry · Statistics · Calculus

## Also included

- Times tables revision with an optional timed check.
- 11+ practice papers and a finder for local test details, covering 33 areas.
- An AI tutor for children and an assistant for parents, both optional.
- Money topics from Year 3, with real-life contexts.
- Read-aloud, sound and motion controls, and light, dark or automatic themes.

YEARS 3 TO 6 · AGES 7 TO 11

# KS2

## Key Stage 2

The foundation. Number sense, the four operations, fractions, measures and shape, built up year by year with short practice, picture models and plenty of coming back to old ideas.

Number | Ratio and proportion | Algebra | Measures | Geometry | Statistics

3 4 5 6

KEY STAGE 2 · AGE 7 TO 11

# Year 3

16 topics first taught in Year 3. Earlier topics keep coming back in review.

**Adding and subtracting**
Number
Written addition and subtraction with big numbers.
*Ask: How could you check your answer is sensible before you work it out exactly?*

**Comparing fractions**
Number
Compare fractions that have the same denominator.
*Ask: Would the same rule work if the denominators were different?*

**Counting in steps**
Number
Count forwards and backwards in 2s, 3s, 4s, 5s, 8s, 10s, 50s and 100s, and find the missing number.
*Ask: How did you work out the jump between the numbers?*

**Fractions with the same bottom number**
Number
Add and subtract fractions with the same denominator, within one whole.
*Ask: Why does the bottom number stay the same when you add?*

**Place value**
Number
The value of each digit, up to hundreds of thousands.
*Ask: Why does a zero matter in 4,073?*

**Times tables**
Number
Multiplication and division facts to 12 × 12.
*Ask: If you know 6 × 8, how can you quickly work out 7 × 8?*

**Adding and taking away lengths and amounts**
Measures
Add and subtract lengths (m, cm, mm), mass (kg, g) and volume or capacity (l, ml).
*Ask: Which unit did you change to, and why?*

**Coins and notes**
Measures
Count amounts made from coins, and make an amount with the fewest coins.
*Ask: Which coins would you use first to make 87p, and why?*

**How much time has passed?**
Measures
Find the time between two clock readings.
*Ask: Why do we count to 60 minutes instead of 100?*

**Money and change**
Measures
Add prices and work out change in pence.
*Ask: How could you count up to the amount paid?*

KEY STAGE 2

# Year 3 (continued)

**Perimeter of shapes**

Measures

Measure and calculate the perimeter of simple shapes.

*Ask: How can you be sure you have counted every side?*

**Reading analogue clocks**

Measures

Tell the time from the hands of a clock, to the nearest five minutes, and change to 24-hour time.

*Ask: Where is the long hand at quarter past, and why is it there?*

**Time facts**

Measures

Know the number of seconds in a minute, hours in a day, days in a week, month or year, and in a leap year.

*Ask: How could you remember which months have 31 days?*

**Parallel, perpendicular, horizontal and vertical lines**

Geometry

Name horizontal, vertical, parallel and perpendicular lines.

*Ask: Where can you spot parallel lines on your way to school?*

**Right angles and turns**

Geometry

Recognise right angles, and describe quarter, half and three-quarter turns.

*Ask: How many right angles are in a half turn, and how do you know?*

**Pictograms**

Statistics

Read and answer questions about pictograms where each symbol stands for more than one.

*Ask: What does one symbol stand for, and how did that help?*

KEY STAGE 2 · AGE 7 TO 11

# Year 4

23 topics first taught in Year 4. Earlier topics keep coming back in review.

**Counting in 6, 7, 9, 25 and 1000**
Number

Count forwards and backwards in multiples of 6, 7, 9, 25 and 1000.
*Ask: How can counting in 9s be done by counting in 10s and taking one off?*

**Counting through zero**
Number

Count backwards through zero into negative numbers, in temperatures and on number lines.
*Ask: How far did you go down to reach zero, and how much further was left?*

**Dividing**
Number

Short and long division, with remainders.
*Ask: What does 'remainder 3' mean in a real situation, like sharing sweets?*

**Dividing by 10 and 100**
Number

Divide whole numbers by 10 and 100, giving tenths and hundredths.
*Ask: Which way do the digits move when you divide, and by how many places?*

**Equivalent fractions**
Number

Fractions that are the same size, and simplifying.
*Ask: Why is 2/4 the same amount as 1/2?*

**Multiplying**
Number

Multiply 2- and 3-digit numbers.
*Ask: How would you split 23 × 14 into easier multiplications?*

**Multiplying three numbers**
Number

Multiply three numbers together, choosing the easiest order.
*Ask: Which two numbers would you multiply first, and why?*

**Roman numerals**
Number

Read Roman numerals to 100, and then to 1,000.
*Ask: Why is IX 9 and not 11?*

**Rounding**
Number

Round whole numbers and decimals.
*Ask: Where on a number line is 350, and why does it round up?*

**Tenths and hundredths**
Number

Connect fractions over 10 and 100 with decimal notation.
*Ask: What does the zero mean in 0.07?*

KEY STAGE 2

# Year 4 (continued)

### Adding and subtracting money

Measures

Add and subtract amounts in pounds and pence, and work out change.

*Ask: How could you count up to work out the change?*

### Area by counting squares

Measures

Find the area of rectangles and simple compound shapes by counting or multiplying squares.

*Ask: Why does multiplying the rows by the columns give the same as counting every square?*

### Converting units of time

Measures

Convert between seconds, minutes, hours, days and weeks.

*Ask: Do you multiply or divide to change hours into minutes, and why?*

### Perimeter of compound shapes

Measures

Find the perimeter of rectilinear shapes, such as an L-shape.

*Ask: Which sides are not labelled, and how could you work out their length?*

### Pounds and pence

Measures

Write amounts of money in pounds and in pence, using decimals.

*Ask: Why is £3.05 not the same as £3.50?*

### Reading scales

Measures

Work out what each mark on a scale is worth, then read the value the pointer shows.

*Ask: How can you tell what each little mark is worth when only some marks have numbers?*

### Timetables and journey times

Measures

Read a timetable and work out how long a journey takes, in minutes.

*Ask: Why is it handy to count up to the next whole hour first?*

### Acute, right, obtuse and reflex angles

Geometry

Name angles as acute, right, obtuse, straight or reflex, and compare their sizes.

*Ask: How could you tell an obtuse angle from an acute one without measuring?*

### Comparing and classifying shapes

Geometry

Name triangles and quadrilaterals from their sides and angles.

*Ask: How is a square different from a rectangle that is not a square?*

### Lines of symmetry

Geometry

Find and count the lines of symmetry in 2-D shapes and letters.

*Ask: How can you test whether a line really is a line of symmetry?*

### Moving on a grid

Geometry

Move points left and right on a coordinate grid.

*Ask: What would change if you moved up instead?*

KEY STAGE 2

# Year 4 (continued)

### Reading bar charts

Statistics

Use the numbers on the left to read and compare bars.

*Ask: How many votes does each gap between the lines stand for?*

### Reading tables and lists

Statistics

Solve sum and difference problems using data in tables and lists.

*Ask: What does each number in the table stand for?*

KEY STAGE 2 · AGE 7 TO 11

# Year 5

20 topics first taught in Year 5. Earlier topics keep coming back in review.

**Decimals**
Number

Add, subtract, multiply and divide decimals.
*Ask: What happens to the digits when you multiply 3.5 by 10?*

**Factors and multiples**
Number

Factors, multiples, highest common factor and lowest common multiple.
*Ask: How can you tell quickly whether a number is a multiple of 3?*

**Fractions of amounts**
Number

Find a fraction of a number or quantity.
*Ask: What does the bottom number tell you, and what does the top number tell you?*

**Mixed numbers and improper fractions**
Number

Count the fraction pieces in whole numbers and mixed numbers.
*Ask: Why does the denominator stay the same?*

**Multiplying fractions by whole numbers**
Number

Multiply a fraction by a whole number, and write the answer as a mixed number.
*Ask: Why does the bottom number not change when you multiply?*

**Numbers to a million**
Number

Read, write and compare numbers to 1,000,000, and multiply and divide by 10, 100 and 1,000.
*Ask: What happens to the digits when you multiply by 1,000?*

**Prime, square and cube numbers**
Number

Know prime numbers up to 19, and use square and cube numbers.
*Ask: How can you tell that 21 is not a prime number?*

**Thousandths**
Number

Read, write and compare decimals with three decimal places, and relate thousandths to tenths and hundredths.
*Ask: Why is 0.4 the same as 0.400?*

**Converting units**
Measures

Convert between metric units of length, mass and capacity.
*Ask: Should the number get bigger or smaller when you change kilograms into grams? Why?*

KEY STAGE 2

# Year 5 (continued)

**Metric and imperial units**
Measures

Convert between common metric and imperial units using a given rule, such as inches and centimetres, or miles and kilometres.
*Ask: How many lots of the rule did you need?*

**Multiplying and dividing money**
Measures

Multiply a price by a number, and share an amount to find the price of one.
*Ask: Would it be easier to work in pounds or in pence here?*

**Perimeter and area**
Measures

Perimeter and area of rectangles and triangles.
*Ask: Can two shapes have the same area but different perimeters?*

**Shopping bills**
Measures

Work out the cost of several items, and the change from a bigger note.
*Ask: What is a quick estimate of the bill before you work it out exactly?*

**Volume of cuboids**
Measures

Use layers of cubes to calculate volume.
*Ask: What does one cubic centimetre look like?*

**3-D shapes**
Geometry

Count the faces, edges and vertices of 3-D shapes.
*Ask: How can you keep track so you don't count an edge twice?*

**Angles**
Geometry

Angles on a line, around a point, in triangles and quadrilaterals.
*Ask: Why do the angles in any triangle always add up to 180°?*

**Nets and 3D solids**
Geometry

Count the faces, edges and corners of a solid, and use them to check a net.
*Ask: Which solids have the same number of faces but different numbers of corners?*

**Properties of rectangles**
Geometry

Use the properties of rectangles to find missing lengths and angles.
*Ask: If you know two sides of a rectangle, how many more sides do you know?*

**Regular and irregular polygons**
Geometry

Tell regular polygons (equal sides and angles) from irregular ones.
*Ask: Why is a rectangle that is not a square an irregular polygon?*

**Line graphs**
Statistics

Read values from a line graph, find changes, and estimate between two readings.
*Ask: What does a steep line tell you compared with a flat line?*

KEY STAGE 2 · AGE 7 TO 11

# Year 6

28 topics first taught in Year 6. Earlier topics keep coming back in review.

**Adding fractions**
Number
Add fractions with different denominators.
*Ask: Why can't we just add the bottom numbers together?*

**Dividing fractions by whole numbers**
Number
Divide a proper fraction by a whole number.
*Ask: Why does the bottom number get bigger when you divide?*

**Fractions, decimals and percentages**
Number
Change between fractions, decimals and percentages.
*Ask: How could you turn a fraction into a percentage?*

**Long division**
Number
Divide numbers up to four digits by a two-digit number, and show remainders as whole numbers, fractions or decimals.
*Ask: How can you check a long division answer by multiplying?*

**Multi-step word problems**
Number
Plan and solve problems with more than one step.
*Ask: Can you explain the first step before you do any calculating?*

**Multiplying fractions**
Number
Multiply proper fractions together.
*Ask: Why is a half of a half a quarter?*

**Negative numbers**
Number
Counting and calculating across zero.
*Ask: How would you explain how far it is from -4 to 5?*

**Numbers up to ten million**
Number
Read, write, order and compare numbers up to 10,000,000 and find the value of each digit.
*Ask: How do the commas help you read a seven-digit number?*

**Order of operations**
Number
BIDMAS with brackets and powers.
*Ask: Why is 2 + 3 × 4 not 20?*

**Percentages of amounts**
Number
Find percentages, then increase and decrease amounts.
*Ask: A shop takes 20% off £60. How would you work out the sale price without a calculator?*

KEY STAGE 2

# Year 6 (continued)

**Scale factors and maps**
Ratio and proportion
Use scale factors to enlarge shapes, and use map scales.
*Ask: What do you multiply by to go from the small shape to the big one?*

**Scaling recipes**
Ratio and proportion
Scale an ingredient to make a different number of servings.
*Ask: How could you check your answer is the right size?*

**Sharing in a ratio**
Ratio and proportion
Split an amount in a given ratio.
*Ask: If two people share in the ratio 3:2, what fraction of the money does each get?*

**Missing number equations**
Algebra
Find the missing number in a one-step or two-step equation.
*Ask: What is the opposite of each step, and in what order do you undo them?*

**Number sequences**
Algebra
Continue sequences and use a rule to find any term.
*Ask: How would you find the 100th term without writing them all out?*

**Two unknowns**
Algebra
Find pairs of numbers that satisfy two conditions, and test possibilities.
*Ask: How can you check your pair against both clues?*

**Using a formula**
Algebra
Use a simple formula for the perimeter of a rectangle.
*Ask: How would you find the width if you knew the perimeter and length?*

**Area of a parallelogram**
Measures
Find the area of a parallelogram using its base and its perpendicular height.
*Ask: Why do we use the straight-up height, not the slanted side?*

**Best value**
Measures
Compare prices to find the best value, by working out the cost of one.
*Ask: How can you compare two packs of different sizes fairly?*

**Changing currency**
Measures
Convert between pounds and other currencies using an exchange rate.
*Ask: Do you multiply or divide to turn euros back into pounds?*

**Same area, different perimeter**
Measures
See that shapes with the same area can have different perimeters, and the other way round.
*Ask: Which 12-tile rectangle has the shortest perimeter, and why?*

**Coordinates in all four quadrants**
Geometry
Use coordinates with negative numbers, and find distances and areas on the full grid.
*Ask: How do you find the distance between –3 and 4 on a number line?*

KEY STAGE 2

# Year 6 (continued)

**Parts of a circle**
Geometry
Use the radius and the diameter of a circle.
*Ask: How are the radius and the diameter linked?*

**Reflecting and translating points**
Geometry
Find the new coordinates of a point after a translation or a reflection.
*Ask: Does a reflection change how far the point is from the mirror line?*

**Vertically opposite angles**
Geometry
Find missing angles where two straight lines cross.
*Ask: Why must the angles opposite each other be the same size?*

**Averages and range**
Statistics
Mean, median, mode and range.
*Ask: When would the median tell you more than the mean?*

**Drawing pie chart angles**
Statistics
Work out the angle for each slice of a pie chart, and use angles to find amounts.
*Ask: What angle does a single person get, and how did you find it?*

**Interpreting pie charts**
Statistics
Use a sector's share of a whole circle to find a count.
*Ask: What would a semicircle tell you?*

YEARS 7 TO 9 · AGES 11 TO 14

# KS3

## Key Stage 3

Where arithmetic turns into algebra. Ratio and proportion, geometry, probability and the ideas GCSE rests on, with every method shown step by step.

Number | Ratio and proportion | Algebra | Measures | Geometry | Statistics

7 8 9

KEY STAGE 3 · AGE 11 TO 14

# Year 7

12 topics first taught in Year 7. Earlier topics keep coming back in review.

**Adding and subtracting mixed numbers**
Number

Add and subtract mixed numbers, giving the answer as an improper fraction.
*Ask: Would it be easier to add the whole numbers and the fractions separately, or to change both to improper fractions?*

**Calculating with negative numbers**
Number

Add, take away, multiply and divide with negative numbers.
*Ask: What is 5 - (-2) on a number line, and why does taking away a negative go up?*

**Prime factors**
Number

Break a number into prime factors, and read powers of primes.
*Ask: Why do we start by dividing by 2, then 3, then 5?*

**Profit and loss**
Number

Work out profit, loss and percentage profit.
*Ask: Why do we divide by the cost price, not the selling price?*

**Significant figures and estimating**
Number

Round to significant figures, and estimate a calculation by rounding.
*Ask: How could an estimate tell you that you pressed the wrong button on a calculator?*

**Proportion problems**
Ratio and proportion

Use the unitary method for direct proportion, and halve the time when you double the workers.
*Ask: How can you check whether your answer is sensible?*

**Ratio problems**
Ratio and proportion

Solve ratio problems when you are given one share or the difference between two shares.
*Ask: How do you find the value of one part when you know one share?*

**Simplifying expressions**
Algebra

Collect like terms.
*Ask: Why can we simplify 3x + 5x but not 3x + 5?*

**Substituting into expressions**
Algebra

Replace a letter with its value and calculate.
*Ask: Why does 4x mean four times x?*

KEY STAGE 3

# Year 7 (continued)

**Area of compound shapes**

Measures

Split a compound shape into rectangles and triangles, and add or subtract their areas.

*Ask: Is there more than one way to split this shape?*

**Area of triangles and trapeziums**

Measures

Find the area of a triangle, parallelogram or trapezium.

*Ask: Why is the area of a triangle half of base times height?*

**Simple probability**

Statistics

Work out the chance of an event with equally likely outcomes.

*Ask: How would adding another blue counter change the probability?*

KEY STAGE 3 · AGE 11 TO 14

# Year 8

15 topics first taught in Year 8. Earlier topics keep coming back in review.

| | |
|---|---|
| **Percentage increase and decrease**<br>Number | Increase or decrease an amount by a percentage, and find a percentage change.<br>*Ask: Why is a 10% rise followed by a 10% fall not back where you started?* |
| **Simple interest**<br>Number | Work out simple interest and the total in an account.<br>*Ask: How is simple interest different from the interest on a savings account that grows each year?* |
| **VAT, tax and wages**<br>Number | Add VAT to a price, take it off, and work out pay with overtime.<br>*Ask: Why is taking 20% off the price with VAT not the same as the price without VAT?* |
| **Expanding brackets**<br>Algebra | Multiply out a single bracket.<br>*Ask: What does 3(x + 4) mean in words, and what are its two parts?* |
| **Factorising with a common factor**<br>Algebra | Factorise expressions by taking out the highest common factor.<br>*Ask: How can you check your factorised answer is right?* |
| **Solving equations**<br>Algebra | Solve linear equations, including unknowns on both sides.<br>*Ask: How can you check your answer is right without looking at a mark scheme?* |
| **Solving inequalities**<br>Algebra | Solve linear inequalities and find the largest or smallest whole number that works.<br>*Ask: Why does the sign flip when you divide by a negative number?* |
| **Substituting negative numbers**<br>Algebra | Substitute negative numbers into expressions and formulae.<br>*Ask: Why does $(-3)^2$ give a positive answer?* |
| **The nth term**<br>Algebra | Find the rule for the nth term of a linear sequence, and use it.<br>*Ask: How do you know what number goes in front of n?* |
| **Speed, distance and time**<br>Measures | Use speed = distance ÷ time to find a missing value.<br>*Ask: If you drive at 40 mph for 2 hours, how far have you gone, and how do you know?* |

KEY STAGE 3

# Year 8 (continued)

### Angles in parallel lines

Geometry

Use corresponding, alternate and co-interior angles.

*Ask: Which pairs of angles are equal in a Z shape, and which add to 180° in a C shape?*

### Constructions and loci

Geometry

Know which construction gives a perpendicular bisector or an angle bisector, and describe a locus.

*Ask: Why does a pair of compasses help when a rule says 'the same distance'?*

### Transformations

Geometry

Translate, reflect, rotate and enlarge points on a coordinate grid.

*Ask: How can you check a reflection is the right distance from the mirror line?*

### Averages from tables

Statistics

Find the total frequency and the mean from a frequency table.

*Ask: Why do we multiply the value by the frequency?*

### Listing outcomes

Statistics

List all the possible outcomes of two events, and count the ways of getting a total.

*Ask: How does an organised list make sure you do not miss an outcome?*

KEY STAGE 3 · AGE 11 TO 14

# Year 9

13 topics first taught in Year 9. Earlier topics keep coming back in review.

**Budgets**
Number

Use percentages and fractions of income to plan spending and saving.

*Ask: If your income stays the same, how long would it take to save for something you want?*

**Standard form: first steps**
Number

Write large and small numbers as a number between 1 and 10 times a power of ten, and multiply them.

*Ask: Why is a calculator's 'E' notation just standard form in disguise?*

**Equation of a straight line**
Algebra

Use y = mx + c: find the gradient and intercept, and work out a point on the line.

*Ask: What does c tell you about where the line is on the graph?*

**Gradient of a straight line**
Algebra

Find how much y changes for each unit of x.

*Ask: What does a gradient of zero tell you?*

**Laws of indices**
Algebra

Simplify powers of the same number when multiplying, dividing and raising to a power.

*Ask: Why does 2^3 × 2^2 give 2^5 and not 2^6?*

**Volume of a cylinder**
Measures

Find the volume of a cylinder from its radius and height.

*Ask: Why can you treat a cylinder like a prism?*

**Volume of prisms**
Measures

Find the volume of a prism from its cross-section and length.

*Ask: Why does a prism's volume use the area of its cross-section?*

**Angles in polygons**
Geometry

Interior and exterior angles.

*Ask: Why do exterior angles always add to 360°? (Imagine walking around the shape.)*

**Bearings**
Geometry

Use three-figure bearings, and find the back bearing.

*Ask: Why does the back bearing differ by 180°?*

**Circumference of a circle**
Geometry

Find the distance around a circle using π.

*Ask: How can you check a circumference answer is sensible without a calculator?*

KEY STAGE 3

# Year 9 (continued)

**Pythagoras' theorem**

Geometry

Find missing sides of right-angled triangles.

*Ask: How do you know which side is the hypotenuse? When do you add the squares and when do you subtract?*

**Scatter graphs and correlation**

Statistics

Describe the type of correlation shown by data, and know that correlation is not cause.

*Ask: Can you think of two things that rise together but one does not cause the other?*

**Venn diagrams**

Statistics

Use Venn diagrams to find how many are in both sets, either, or neither.

*Ask: Why do you take off the overlap once when you add two sets?*

YEARS 10 TO 11 · AGES 14 TO 16

# GCSE

GCSE

Foundation and Higher content: algebra, geometry and trigonometry, statistics and probability, and problem solving, with the methods taught before the exam wording.

Number | Ratio and proportion | Algebra | Measures | Geometry | Statistics

10 11

GCSE · AGE 14 TO 16

# Year 10

23 topics first taught in Year 10. Earlier topics keep coming back in review.

**Reverse percentages**
Number
Recover an original price after a percentage change.
*Ask: Why does adding 20% back to a sale price give the wrong answer?*

**Simplifying surds**
Number
Simplify surds by taking out square factors, and add them.
*Ask: Which square numbers should you look for first?*

**Standard form**
Number
Interpret powers of ten in standard form.
*Ask: What changes when the power of ten is negative?*

**Direct and inverse proportion**
Ratio and proportion
Use $y = kx$ and $y = k/x$ to solve proportion problems.
*Ask: How do you know whether two things are in direct or inverse proportion?*

**Completing the square**
Algebra
Write $x^2 + bx + c$ as $(x + p)^2 + q$, and find the minimum point.
*Ask: Why is the minimum value the number outside the bracket?*

**Expanding double brackets**
Algebra
Expand and simplify pairs of brackets.
*Ask: Draw a rectangle with sides $(x + 3)$ and $(x + 2)$. Can you see where each of the four terms comes from?*

**Factorising quadratics**
Algebra
Write quadratics as pairs of brackets.
*Ask: How could you check a factorisation using expanding?*

**Quadratic sequences**
Algebra
Use the second difference to spot a quadratic sequence, and find later terms.
*Ask: Why is the second difference always twice the coefficient of $n^2$?*

**Rearranging formulae**
Algebra
Change the subject of a formula, and use it to find a missing value.
*Ask: Which operation do you undo first, and why?*

**Simplifying algebraic fractions**
Algebra
Simplify algebraic fractions by factorising and cancelling.
*Ask: Why can you cancel a factor but not a single term?*

GCSE

# Year 10 (continued)

**Simultaneous equations**
Algebra
Solve two equations with two unknowns.
*Ask: What does the solution mean if you draw both equations as lines on a graph?*

**Solving quadratics**
Algebra
Solve quadratic equations by factorising.
*Ask: Why can a quadratic have two answers?*

**The quadratic formula**
Algebra
Use the discriminant, and solve quadratics with the formula.
*Ask: What does it tell you if the discriminant is negative?*

**Density and pressure**
Measures
Use density = mass ÷ volume and pressure = force ÷ area.
*Ask: Why does a high heel press harder on the floor than a flat shoe, if the force is the same?*

**Volume of cones and spheres**
Measures
Use the formulae for the volume of a cone and a sphere.
*Ask: How is the volume of a cone related to the volume of a cylinder?*

**Arc length and sector area**
Geometry
Find the length of an arc and the area of a sector.
*Ask: What fraction of the circle is a 90° sector?*

**Area of a circle**
Geometry
Use radius or diameter to calculate the area of a circle.
*Ask: What happens to the area when the radius doubles?*

**Circle theorems**
Geometry
Use the angle at the centre, angles in a semicircle, cyclic quadrilaterals and tangents.
*Ask: Which circle theorem did you use, and how did you know?*

**Exact trigonometric values**
Geometry
Know the exact values of sin, cos and tan for 0°, 30°, 45°, 60° and 90°.
*Ask: How can a half equilateral triangle help you remember the values?*

**Right-angled trigonometry**
Geometry
Use sin, cos and tan to find sides and angles.
*Ask: How do you decide whether to use sin, cos or tan?*

**Sine and cosine rules**
Geometry
Use the sine rule, the cosine rule and ½ab sin C in any triangle.
*Ask: How do you decide between the sine rule and the cosine rule?*

GCSE

# Year 10 (continued)

| | |
|---|---|
| **Vectors**<br>Geometry | Add and scale column vectors, and find the length of a vector.<br>*Ask: How can you draw a vector on a grid?* |
| **Histograms and frequency density**<br>Statistics | Use frequency density = frequency ÷ class width to read and draw histograms.<br>*Ask: Why do we use frequency density instead of frequency for the height?* |

GCSE · AGE 14 TO 16

# Year 11

14 topics first taught in Year 11. Earlier topics keep coming back in review.

| | |
|---|---|
| **Compound interest**<br>Number | Work out an amount after interest is added each year.<br>*Ask: Why does money grow faster with compound interest than with simple interest?* |
| **Growth and decay**<br>Number | Use multipliers for repeated percentage growth and decay.<br>*Ask: What is the multiplier for a 12% decrease?* |
| **Recurring decimals as fractions**<br>Number | Turn a recurring decimal into a fraction.<br>*Ask: Why does subtracting make the repeating part disappear?* |
| **Upper and lower bounds**<br>Number | Find the largest and smallest values a rounded number could have been.<br>*Ask: What is the smallest number that rounds to 8 when rounded to the nearest whole number?* |
| **Composite functions**<br>Algebra | Put one function inside another.<br>*Ask: Does f(g(2)) always equal g(f(2))? Try it.* |
| **Inverse functions**<br>Algebra | Find the inverse of a function and use it.<br>*Ask: How can you check that you have the right inverse?* |
| **Iteration**<br>Algebra | Use an iterative formula to find the next terms.<br>*Ask: How do you use the answer from one step in the next one?* |
| **Quadratic graphs**<br>Algebra | Find the turning point, intercepts and symmetry of a quadratic graph.<br>*Ask: Where is the turning point, compared with the two roots?* |
| **Transforming graphs**<br>Algebra | Describe how y = f(x) moves, stretches or reflects when the equation changes.<br>*Ask: Why does f(x + 3) move the graph left, not right?* |
| **Graphs of sine and cosine**<br>Geometry | Use the shape of sin and cos graphs: amplitude, maximum, minimum and period.<br>*Ask: What happens to the wave if you multiply x by 2 inside the brackets?* |

GCSE

# Year 11 (continued)

**Similar shapes and scale factors**

Geometry

Use a scale factor to find missing lengths and areas.

*Ask: If every length doubles, what happens to the area?*

**Conditional probability**

Statistics

Find probabilities when items are taken without replacement.

*Ask: Why do the bottom numbers go down after the first pick?*

**Cumulative frequency and box plots**

Statistics

Find the median and quartiles from a cumulative frequency graph, and the interquartile range.

*Ask: What does a small interquartile range tell you about the data?*

**Probability of two events**

Statistics

Combine the probabilities of two independent events.

*Ask: Why do we multiply along the branches of a tree diagram?*

YEARS 12 TO 13 · AGES 16 TO 18



# A-level

Pure maths, statistics and mechanics: calculus, sequences and series, proof, distributions and modelling, with accurate notation and justified answers.

Algebra · Measures · Geometry · Statistics · Calculus

12 13

A-LEVEL · AGE 16 TO 18

# Year 12

22 topics first taught in Year 12. Earlier topics keep coming back in review.

**Arithmetic sequences and series**
Algebra

Find a term, or the sum of the first terms, of an arithmetic sequence.
*Ask: Why does pairing the first and last terms give a quick way to add up an arithmetic sequence?*

**Exponential growth and logarithms**
Algebra

Use exponential models, and logarithms to solve for the time.
*Ask: What does taking a logarithm do to a power?*

**Logarithms and their laws**
Algebra

Evaluate logarithms and combine them using the laws.
*Ask: What does log_2 32 mean in words?*

**Partial fractions**
Algebra

Split an algebraic fraction into two simpler fractions.
*Ask: Why does putting x equal to a root make one of the terms disappear?*

**Proof by counter-example**
Algebra

Disprove a statement with a counter-example.
*Ask: Why does one counter-example disprove a statement, but many examples do not prove it?*

**Sigma notation**
Algebra

Evaluate sums written with sigma notation, using the standard results.
*Ask: How could you check a sum of 5 terms without a formula?*

**The factor theorem and remainders**
Algebra

Use f(a) to find remainders and factors of polynomials.
*Ask: What does it tell you if f(a) = 0?*

**The modulus function**
Algebra

Evaluate and solve equations and inequalities with a modulus.
*Ask: Why does an equation with a modulus often have two solutions?*

**Motion with constant acceleration**
Measures

Choose and use a constant-acceleration equation.
*Ask: Why does the equation depend on acceleration being constant?*

A-LEVEL

# Year 12 (continued)

**Radians**
Geometry
Convert between degrees and radians, and find arc lengths and sector areas in radians.
*Ask: How many degrees is one radian roughly?*

**Solving trigonometric equations**
Geometry
Solve simple trigonometric equations for angles between 0° and 360°.
*Ask: Why are there two solutions between 0° and 360°?*

**Trigonometric identities**
Geometry
Use $\sin^2\theta + \cos^2\theta = 1$ and $\tan\theta = \sin\theta \div \cos\theta$ to find missing values.
*Ask: Why is $\sin^2\theta + \cos^2\theta = 1$ just Pythagoras' theorem in disguise?*

**Hypothesis testing: the ideas**
Statistics
Write hypotheses, say whether a test is one-tailed or two-tailed, and decide whether to reject $H_0$.
*Ask: Why do we say 'not enough evidence to reject $H_0$' rather than '$H_0$ is true'?*

**Regression lines and residuals**
Statistics
Use a regression line to predict, read its gradient in context, and find a residual.
*Ask: Why is a prediction outside the range of the data less trustworthy?*

**Sampling methods**
Statistics
Name a sampling method from how it works, and spot where it could be biased.
*Ask: Why might asking only your friends give a biased result?*

**The binomial distribution**
Statistics
Use $X \sim B(n, p)$: the mean, the variance and the probability of exactly k successes.
*Ask: What conditions must be true for a binomial model?*

**The normal distribution**
Statistics
Use the mean, standard deviation, z-scores and the 68-95-99.7 rule.
*Ask: What does a z-score of −1 tell you?*

**Definite integrals of linear functions**
Calculus
Find an antiderivative and evaluate it at two limits.
*Ask: Why does the constant of integration cancel?*

**Differentiating polynomials**
Calculus
Use the power rule to find gradients.
*Ask: What does dy/dx tell you about the graph of y?*

**Finding stationary points**
Calculus
Connect a zero derivative with a horizontal tangent.
*Ask: Why does a zero derivative alone not establish a minimum?*

A-LEVEL

# Year 12 (continued)

**Integrating polynomials**

Calculus

Integrate powers of x, and evaluate definite integrals, including areas under curves.

*Ask: How is integration the reverse of differentiation?*

**The product and quotient rules**

Calculus

Differentiate products and quotients, and find the gradient at a point.

*Ask: Which part is u and which is v, and how did you choose?*

A-LEVEL · AGE 16 TO 18

# Year 13

13 topics first taught in Year 13. Earlier topics keep coming back in review.

**Binomial expansion coefficients**
Algebra

Find the coefficient of a term when a bracket is raised to a power.
*Ask: How does Pascal's triangle give the numbers in front of each term?*

**Parametric equations**
Algebra

Find points on a curve from x(t) and y(t), and find dy/dx using the chain rule.
*Ask: Why is dy/dx the same as dy/dt divided by dx/dt?*

**Sum to infinity of a geometric series**
Algebra

Find what a geometric series adds up to when it goes on for ever.
*Ask: Why does a series with ratio 2 have no sum to infinity?*

**Forces and moments**
Measures

Resolve forces, use F = ma, and balance moments.
*Ask: Which direction does each force turn the beam?*

**Projectile motion**
Measures

Split projectile motion into horizontal and vertical parts, with g = 9.8 m $s^{-2}$.
*Ask: Why can you treat the horizontal and vertical motion separately?*

**Vectors in three dimensions**
Geometry

Work with 3-D vectors: their length, the scalar product and the distance between points.
*Ask: What does it tell you if the scalar product of two vectors is zero?*

**Differential equations**
Calculus

Solve simple differential equations by integrating, using a given point.
*Ask: How do you find the value of the constant after integrating?*

**Differentiating exponentials and trigonometric functions**
Calculus

Differentiate $e^{kx}$, sin kx, cos kx and ln x, and find the gradient at a point.
*Ask: What does the number in front of x do to the derivative?*

**Differentiating with the chain rule**
Calculus

Differentiate a function of a function, such as (2x+1)^3.
*Ask: Why do we multiply by the derivative of the inside part?*

A-LEVEL

# Year 13 (continued)

**Integration by substitution**
Calculus
Integrate a function when part of it is the derivative of another part.
*Ask: How can you tell which part to call u?*

**Kinematics using calculus**
Calculus
Differentiate displacement to find velocity and acceleration.
*Ask: What does it mean if velocity is zero at a moment?*

**Newton-Raphson iteration**
Calculus
Use Newton-Raphson to improve an estimate of a root.
*Ask: Why does the formula use the gradient of the curve?*

**The trapezium rule**
Calculus
Estimate the area under a curve by splitting it into trapeziums.
*Ask: Would more strips make the estimate better or worse, and why?*

YEARS 4 TO 6

# 11+

# Preparing for selective schools

Maths topics at the right level, verbal and non-verbal reasoning, problem solving, practice papers and a finder for what your local test involves.

SELECTIVE SCHOOL PREPARATION

# What's in the 11+ strand

## Verbal reasoning

Word and letter patterns, codes, analogies, odd words out and hidden words.

## Non-verbal reasoning

Shapes and pictures: series, matrices, reflections, rotations and odd one out.

## Problem solving

Logic puzzles, finding the error, combinations and “always, sometimes, never” reasoning.

## Practice papers

**Maths paper**
20 questions · 25 minutes

**Reasoning paper**
Verbal, non-verbal and problem solving · 20 questions · 20 minutes

**Maths and reasoning**
Two parts in one sitting · 30 questions · 35 minutes

Papers are timed, with no hints, and marks are shown at the end with the topics to revisit. They're the one place quarter uses a countdown.

## Local test finder

Tests differ between areas and schools. The finder in the parents' guides covers 33 areas of England and Northern Ireland: search by postcode, area or school, see what's known about each test and follow the links to the official admissions pages.

START EARLY AND KEEP IT LIGHT. TWENTY MINUTES A FEW TIMES A WEEK BEATS A LAST-MINUTE RUSH.

THE 11+

# Verbal reasoning

**Odd one out (words)**
Reasoning
Find the word that does not belong with the others.
*Ask: Say four things that are fruit and one that is not. How did you decide?*

**Who is first? (ordering clues)**
Reasoning
Put people in order from clues like 'Ben is taller than Cara'.
*Ask: Make up three clues about your family's heights. Can someone work out who is tallest?*

**Letter series**
Reasoning
Spot the pattern in a row of letters and find the next one.
*Ask: What comes next in A, C, E, G? How many places does each letter move?*

**Alphabet codes**
Reasoning
Work out a secret code and use it on a new word.
*Ask: If A becomes C, what does B become? What would Z become?*

**Number series**
Reasoning
Find the rule in a row of numbers and carry it on.
*Ask: What comes next in 2, 4, 8, 16? How do you know it is not 2, 4, 8, 14?*

**Word links**
Reasoning
Cat is to kitten as dog is to...? Find the link and use it again.
*Ask: Can you make up a word link of your own for someone else to solve?*

**Odd one out (numbers)**
Reasoning
Find the number that does not belong in the list.
*Ask: Which numbers in 6, 12, 18, 25, 30 are in the 6 times table? Which one is not?*

**Rearrange the letters**
Reasoning
Rearrange scrambled letters to make a word.
*Ask: Can you scramble a word for someone else to unscramble?*

**Hidden words**
Reasoning
Find the word hidden where two words join.
*Ask: Which letters come from the first word and which from the second?*

**Words that go together**
Reasoning
Find the word that goes in front of three others.
*Ask: Can you think of another word that goes in front of 'light'?*

**Letter bridge**
Reasoning
Find the one letter that finishes one word and starts the next.
*Ask: How do you know when a letter works for the first word but not the second?*

THE 11+

# Non-verbal reasoning

**Odd one out (shapes)**

Reasoning

Find the picture that is different from the rest.

*Ask: Show me which pictures match. What is different about the odd one?*

**Shape series**

Reasoning

See how a shape changes and draw the next one.

*Ask: What changes from the first picture to the second? Does it change the same way next?*

**Mirror images**

Reasoning

Find the reflection of a pattern in a mirror line.

*Ask: If you hold a mirror here, what do you see? Which square is closest to the mirror?*

**Quarter turns**

Reasoning

Turn a pattern a quarter turn and find the new picture.

*Ask: Which way is clockwise? Where does the top row go after a quarter turn?*

**Picture links**

Reasoning

The first picture changes into the second. Do the same to the third.

*Ask: What happened to the first shape? Do you need to do more than one thing?*

**Grid puzzles**

Reasoning

Work out the missing picture in a 3 by 3 grid.

*Ask: Which shapes are in the top row? Which one is missing from the bottom row?*

THE 11+

# Problem solving

**Always, sometimes, never**

Reasoning

Decide whether a statement about numbers is always true.

*Ask: Can you find an example that proves 'all prime numbers are odd' is not always true?*

**Spot the mistake**

Reasoning

Look at someone's working and find the line that goes wrong.

*Ask: Which line would you check first, and why? Is the mistake in the sum or in the method?*

**Counting possibilities**

Reasoning

Count all the different ways, without missing any or counting twice.

*Ask: How many different outfits could you make from 3 tops and 2 skirts? Can you list them?*

**Logic puzzles**

Reasoning

Use clues to work out who has what.

*Ask: Which clue helped most? What did it let you rule out?*

YEARS 3 TO 13

# STEM

## The STEM lab

Computing, data science, AI and machine learning, and the ideas of quantum physics, with topics for every age and hands-on labs to build and try.

BEYOND MATHS

# How the STEM lab works

The lab sits next to the maths activities, with its own sticky note on Today. It has its own topics and progress, separate from the maths plan, so children can explore for the fun of it.

| | |
|---|---|
| **Computing** | Precise instructions, loops, sorting and searching, binary and hexadecimal, tracing code, logic, storage and secret codes. |
| **Data science** | Reading survey data, spotting outliers and checking whether data is fair. |
| **AI and machine learning** | Learning from examples, bias in data, nearest neighbour, a tiny artificial neuron, accuracy, spam filters, gradient descent and a neural network layer. |
| **Quantum ideas** | Packets of energy, qubits and probability, how many states qubits hold, and the X and H gates. |

## Build it labs

**Binary lamps.** Switch lamps on and off to build numbers.

**Program the robot.** Queue up instructions to reach the star.

**Teach a machine.** Place dots and see how a nearest-neighbour model decides.

**Qubit playground.** Try the X and H gates and measure the qubit.

THE STEM LAB · YEARS 3 TO 6

# Key Stage 2 STEM topics

**Giving a robot precise instructions**

STEM

Follow a list of instructions step by step to see where a robot ends up.

*Ask: What happens if you swap two of the instructions round?*

**Loops: repeating instructions**

STEM

Work out what a repeat loop does, and how many times things happen.

*Ask: Why is writing 'repeat 10 times' better than writing the instruction ten times?*

**Inputs, processes and outputs**

STEM

A program takes an input, does something to it, and gives an output. Work out the output, or the input.

*Ask: If you know the output, how can you work out the input?*

**Sorting step by step**

STEM

Follow a simple sorting algorithm: swap neighbours that are in the wrong order.

*Ask: Why does the biggest number always end up at the right-hand end after one pass?*

**Data detective**

STEM

Read a table of survey results and answer questions about it, like a data scientist.

*Ask: What other question could you answer with this data?*

**Spotting bugs**

STEM

Find what is wrong with a short program and choose the fix.

*Ask: How would you test a program to find out where the bug is?*

**How computers learn from examples**

STEM

Look at labelled examples and find the feature that tells two groups apart, like a machine learning program does.

*Ask: What could go wrong if the examples were all the same kind of thing?*

THE STEM LAB · YEARS 7 TO 9

# Key Stage 3 STEM topics

**Binary numbers**
STEM
Convert between binary and ordinary (denary) numbers, and add in binary.
*Ask: Why do computers use only 0 and 1?*

**Data types**
STEM
Name the data type of a value: whole number, decimal, text or True/False.
*Ask: Why does a computer need to know what type a value is?*

**AND, OR and NOT**
STEM
Work out whether a logic statement is True or False.
*Ask: Can you think of a rule in real life that uses AND, and one that uses OR?*

**Hexadecimal and colours**
STEM
Read hexadecimal (base 16) numbers, as used for colours on screens.
*Ask: Why is hex handy for programmers who work with binary?*

**Odd one out in data**
STEM
Spot an outlier in a data set and see how much it changes the mean.
*Ask: Would you always remove an outlier? What if it's real?*

**Tracing code**
STEM
Follow a short piece of code line by line and work out the final value of a variable.
*Ask: What does x = x + 1 mean, when x can't be equal to x + 1?*

**A tiny artificial neuron**
STEM
Work out the output of a simple artificial neuron: multiply the inputs by weights, add them up, and compare.
*Ask: If a weight is bigger, what does that tell us about that input?*

**How good is the model?**
STEM
Work out how accurate a machine learning model is from its right and wrong answers.
*Ask: Why shouldn't we test a model on the examples it learned from?*

**Is the data fair?**
STEM
Spot when data or a sample is unfair, and see why that matters for AI.
*Ask: Who might be missing from the data when you only ask your friends?*

**Nearest neighbour: how a machine decides**
STEM
Use the nearest-neighbour idea to classify something new from labelled examples.
*Ask: What could go wrong if the closest example happens to be a mistake?*

THE STEM LAB · YEARS 10 TO 11

# GCSE STEM topics

**Binary search**

STEM

See how halving the list each time finds an item quickly in a sorted list.

*Ask: Why does binary search only work on a sorted list?*

**How fast does the work grow?**

STEM

Compare how the number of steps in an algorithm grows as the amount of data grows.

*Ask: Which would you rather use on a list with a million items: one that takes n steps, or one that takes n × n?*

**Secret codes: Caesar cipher**

STEM

Encrypt and decrypt messages by shifting letters, and use remainders to wrap around the alphabet.

*Ask: Why is a Caesar cipher easy to crack?*

**Storage and file sizes**

STEM

Calculate how many bits and bytes are needed to store text, pictures and sound.

*Ask: Why does a bigger picture take more storage?*

**Probability and the spam filter**

STEM

Use counts to find the chance that an email is spam, given that it contains a word.

*Ask: Why does a word being common in spam not make every email with it spam?*

**Quantum ideas**

STEM

Meet the big ideas of quantum physics: energy comes in packets, and nature at tiny scales works with probabilities.

*Ask: Why do you think quantum effects aren't noticeable in everyday life?*

THE STEM LAB · YEARS 12 TO 13

# A-level STEM topics

**How many states can qubits hold?**

STEM

See how the number of possible measurement outcomes doubles with each extra qubit.

*Ask: What does doubling again and again do to a number?*

**Information and bits**

STEM

Measure information in bits: how many yes/no questions are needed to pin down one outcome.

*Ask: Why do repeated halvings of the possibilities link to powers of 2?*

**Learning by going downhill**

STEM

Take a step of gradient descent, the method that trains neural networks.

*Ask: What would happen if the step (learning rate) was far too big?*

**Qubits and probability**

STEM

Use the amplitudes of a qubit to find the chances of measuring 0 or 1.

*Ask: Why must the two probabilities add up to 1?*

**A neural network layer**

STEM

Calculate the output of a layer of a neural network using a weights table (matrix), inputs, bias and ReLU.

*Ask: Why might a network need an activation function like ReLU?*

**Quantum gates**

STEM

Follow what the X and H gates do to a qubit, and work out the chance of each result.

*Ask: How is the X gate like the NOT gate from classical logic?*

AROUND THE QUESTIONS

# Tools and extras

## Times tables

A grid that fills in as facts become quick, short rounds that lean on the tricky ones, and an optional timed check. It has its own sticky note on Today.

## Working-out tools

Counters, number lines, grids, abacus and bean-counting tools, and method sheets, beside every question.

## The AI tutor

An optional chat that helps a child think a problem through without handing over the answer. Switch it on or off in the parents' area. Chats stay on the device.

## The parents' assistant

Ask how to explain a topic, what to practise next or how to help at home, in plain language.

## Progress report

Choose any dates with a calendar picker, then print or save what was practised, what's secure and what needs attention.

## Look and feel

Light, dark or automatic themes, read-aloud, sound effects you can switch off, reduced motion, and a layout that fits phones, tablets and computers.

## The grown-ups' area

Protected by a quick maths question and an optional PIN. It holds progress, settings, the tutor switch, feedback and your subscription, and lets you export or delete everything saved under your account.

IDEAS, EXPLAINED

# Study library

Short illustrated guides to the big ideas, written for learners with a section for grown-ups. Find them from any topic or at **yourquarter.co.uk/learn**.

### Algorithms and loops
An algorithm is a list of exact steps, and a loop repeats steps so you do not have to write them again and again.

Years 3 to 9

### Angles in a triangle
The three angles in any triangle always add up to 180°.

Years 5 to 9

### Area of rectangles and triangles
Area is the space inside a shape. A rectangle is length × width, and a triangle is half of that.

Years 4 to 9

### Averages: mean, median, mode and range
Different ways to describe a typical value, and how spread out the data is.

Years 6 to 11

### Binary numbers
Computers count with only 0 and 1, and every number can be built from doubling place values.

Years 5 to 11

### Differentiation and the tangent
The derivative gives the gradient of a curve at any point, which is the gradient of its tangent.

Years 12 and 13

### Equivalent fractions
Different fractions can name the same amount, and multiplying or dividing top and bottom by the same number keeps the size.

Years 3 to 7

### Fair and unfair data
An AI learns from its data, so missing or lopsided data can make it work worse for some people.

Years 7 to 13

### Gradient and straight-line graphs
The gradient of a line tells you how steep it is, and y = mx + c describes it.

Years 8 to 11

### How machines learn from examples
A machine learning program finds patterns in labelled examples, then uses them to decide about something new.

Years 6 to 13

### Multiplication as groups and arrays
Multiplication means equal groups, and a grid of dots shows why the order does not matter.

Years 3 to 6

### Negative numbers and the number line
Numbers below zero live on the left of the number line, and adding and subtracting are moves along it.

Years 4 to 8

STUDY LIBRARY

# Study library (continued)

### Neural networks
An artificial neuron multiplies inputs by weights, adds them up, and fires if the total is big enough. Networks join many together.

Years 9 to 13

### Parts of a circle
The radius, diameter and circumference, and how they are linked.

Years 6 to 11

### Pascal's triangle
A triangle of numbers where each one is the two above it added together, and the rows give the numbers in a binomial expansion.

Years 9 to 13

### Percentages
A percentage is a number out of 100, and it links to fractions and decimals.

Years 5 to 11

### Place value
The position of a digit tells you how much it is worth.

Years 3 to 6

### Powers and indices
A power tells you how many times to multiply a number by itself, and the index laws make working with powers quick.

Years 6 to 10

### Prime numbers and factor trees
Every number can be built from prime numbers, and a factor tree shows how.

Years 5 to 9

### Probability tree diagrams
A tree diagram shows every outcome of two events and how to multiply along the branches.

Years 8 to 13

### Pythagoras' theorem
In a right-angled triangle, the square on the longest side equals the sum of the squares on the other two.

Years 9 to 11

### Quadratic graphs and their roots
A quadratic makes a U-shaped curve, and its roots are where it crosses the x-axis.

Years 10 to 12

### Quantum ideas: packets and probabilities
At the scale of atoms, energy comes in packets, and measurements give results with probabilities.

Years 10 to 13

### Secret codes and ciphers
A Caesar cipher shifts every letter along the alphabet. Remainders make the alphabet wrap round, and modern codes use the same idea with huge numbers.

Years 9 to 13

### Sequences and the nth term
A sequence follows a rule. The nth term is a formula that gives any term from its position.

Years 4 to 10

### Sorting and searching
Computers sort lists by comparing neighbours, and search sorted lists by halving them again and again.

Years 4 to 11

STUDY LIBRARY

# Study library (continued)

### Standard form
Standard form writes very big or very small numbers as a number between 1 and 10 times a power of ten.

Years 9 to 11

### Telling the time
Read an analogue clock: the short hand shows the hour and the long hand shows the minutes.

Years 2 to 4

### Trigonometry: sine, cosine and tangent
In a right-angled triangle, sin, cos and tan link an angle to the ratio of two sides.

Years 9 to 11

PRACTICAL TIPS

# Helping at home

## Before practice

- Pick a calm moment, and stop when focus fades.
- Two or three topics are enough for a sitting.
- Have paper and a pencil to hand.

## During practice

- Let them try first. A hint is always there.
- Ask them to explain a step in their own words.
- Praise the effort and the method, not just the answer.

## When something's tricky

- Use “Show the steps”, then try a fresh question.
- Use a real object: coins, a ruler, slices of fruit.
- Come back to it tomorrow. Topics return on their own.

## After practice

- Look at the “Ask” question for the topic.
- Spot the topic in real life: shopping, cooking, travel.
- Check the parents' area now and then, and celebrate what's secure.

IF A TOPIC FEELS TOO EASY OR TOO HARD, TRY ONE FROM THE YEAR ABOVE OR BELOW. THE MAP IS THERE TO BE EXPLORED.

## Keeping in touch

Questions, ideas and feedback are always welcome through the feedback form in the parents' area, or at **yourquarter.co.uk**.

# quarter™

Maths and STEM, one quarter at a time.



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